SMART INDIA HACKATHON 2026

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PRAMAAN-AI

Provenance-anchored Regulatory AI for Approval, Authentication & Norm-compliance

Problem Statement ID — SIH26261

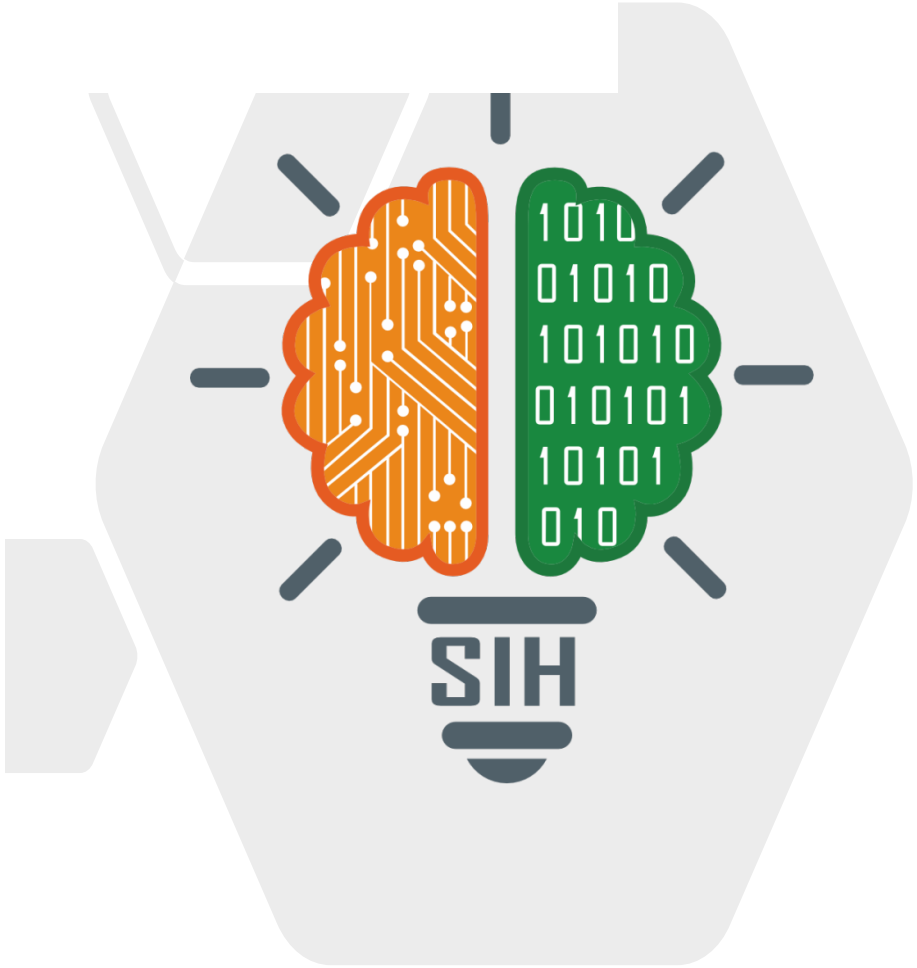
Problem Statement Title — AI-Powered Application Validation, Document Authentication & Regulatory Compliance Verification

Theme — Smart Automation

PS Category — Software

Team ID — CtrlAltElite

Team Name — CtrlAltElite



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PRAMAAN-AI: PROOF, NOT PAPERWORK

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Proposed Solution (Describe your Idea/Solution/Prototype)

- PRAMAAN-AI is an AI pre-scrutiny layer between the AICTE portal and the evaluator's desk.
- PARSE — checks APH completeness, expiry, prescribed formats and arithmetic before review.
- PROVE — uses ELA, metadata, fonts, signatures and copy/splice inspection to surface evidence.
- MATCH + VERIFY + SEE — history/faculty graph, DigiLocker/API Setu/DSC/QR, GPS and photo-reuse checks.
- Innovation — APH-as-Code, claim-level page/region evidence, explainable Trust Score and hard gates.
- AI advises; the authorised human official decides. No black-box verdicts.

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TECHNICAL APPROACH

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- Flow — AICTE portal/upload → PARSE → Claim Graph → five engines → Trust Score → evidence report → human decision/EVC.
- Stack — Python/FastAPI MVP + reviewer dashboard; production seams for PaddleOCR, LayoutLMv3, YOLOv8, OpenCV/ELA, PostgreSQL, Redis and MinIO.
- Method — ingest and classify; extract claims with coordinates; run engines in parallel; score with explainable weights and hard gates; report; learn from human review.
- Working prototype — offline engine, FastAPI application and reviewer dashboard are included with the submission.
- Reproducible demo — clean ELA risk 0.00 vs tampered 0.83; demo Trust Score 52/100 (AMBER); seven-entry SHA-256 audit chain verifies.

FEASIBILITY AND VIABILITY

- Feasible now — open-source technology, commodity CPU/GPU compute, standard issuer interfaces and an API sidecar to the existing portal.
- Scans or regional-language documents → OCR confidence routing; missing labels → synthetic tamper generator plus forensic signals.
- Issuer outage → DSC/QR/manual-verify queue, never automatic rejection; privacy → RBAC, salted graph keys, redaction and on-prem/VPC deployment.
- Adoption — advisory-only rollout keeps official authority; APH-as-Code absorbs handbook changes without a rewrite.
- Roadmap — Phase 0 pilot → Phase 1 shadow mode → Phase 2 assisted live triage → Phase 3 scale and learn.
- Kill criterion — an underperforming engine stays advisory-only until shadow-mode evidence supports it.

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IMPACT AND BENEFITS

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- Design target — approximately 80% less pre-scrutiny effort and first feedback in days rather than weeks.
- 100% evidence-linked — every flag opens the exact source page/region; only RED is prioritised for EVC inspection first.
- Institutions — instant deficiency lists, fewer rejection cycles and predictable cure timelines.
- Evaluators — less page-turning and more judgement; effort shifts to AMBER/RED cases where it matters.
- AICTE and the system — fraud-pattern intelligence, stronger deterrence and fewer unnecessary physical visits or paper files.
- Projected triage mix — 58% GREEN, 29% AMBER, 13% RED. All impact figures are targets to validate in shadow mode.

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RESEARCH AND REFERENCES

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- AICTE Approval Process Handbook 2024–25 to 2026–27 — norms, checklists and EoA process: aicte-india.org.
- DigiLocker Issuer/Requester API specifications and API Setu (NeGD, MeitY) — signed-source verification model.
- LayoutLMv3 (ACM Multimedia 2022), PaddleOCR, OpenCV/ELA and copy-move detection literature — document and image forensics.
- Ultralytics YOLOv8 for infrastructure audit seams; NIST and OWASP guidance for RBAC, audit logging and tamper-evident records.
- Built for this hackathon — offline engine, FastAPI app, reviewer dashboard, Trust Score and hash-chain audit. Prototype data is synthetic; impact figures are targets, not measured claims.
- PRAMAAN-AI does not replace an AICTE official. It removes paperwork standing between officials and their judgement.